

Aurdino based memory read

Abstract

This project, 'Aurdino based memory read', demonstrates a simple Arduino-based system for reading and displaying battery voltage or stored memory values on an LCD. The system uses an Arduino Uno, an LCD (16x2) via I2C or parallel interface, analog input for voltage sensing, and a small supporting circuit (voltage divider and connectors). The aim is to provide a compact educational platform to teach electronics concepts: ADC reading, I2C/LCD interfacing, and basic firmware logic for threshold detection.

Hardware Components

- Arduino Uno (or compatible microcontroller board)
- 16x2 LCD module (with or without I2C backpack)
- Voltage divider (resistors) for scaling battery voltage to Arduino analog input range
- 9V battery and snap connector (or other power source)
- Perforated PCB / protoboard and jumper wires
- Potentiometers (for LCD contrast and other adjustments)
- Optional: level-shifting/protection diodes and connectors

Circuit / Prototype Photos

Below are photos of the assembled prototype and wiring. The first image shows the Arduino, the prototype board with IC and potentiometers, and the 9V battery snap. The second image shows the LCD module used for displaying the readings.

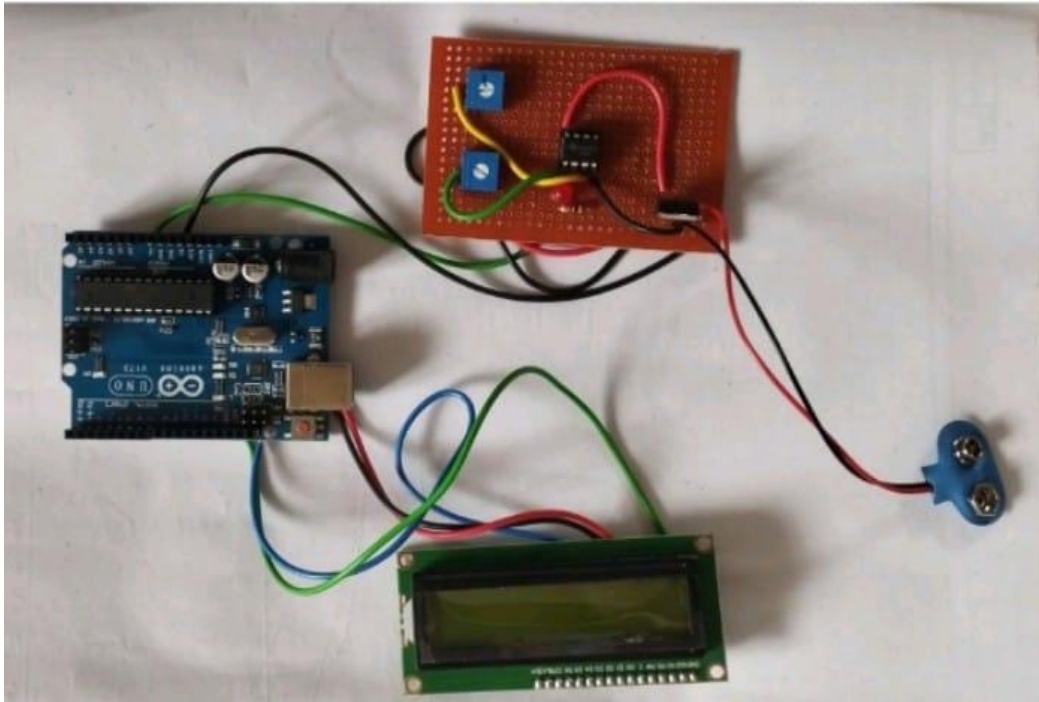


Figure : 16x2 LCD module

3.4 Arduino Code Functionality

```
#include <LCD_I2C.h>
LCD_I2C lcd(0x27, 16, 2);
float input_voltage = 0.0;
void setup()
{
  Serial.begin(9600);
  lcd.begin(); // If you are using more I2C devices using the Wire library use lcd.begin(false)
  lcd.backlight();
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print(" Hello ALL ..."); // You can make spaces using well... spaces
  delay(2000);
}
void loop() {
  int analog_value = analogRead(A0);
  input_voltage = (analog_value * 5.0) / 1024.0;
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Voltage = ");
  lcd.setCursor(10,0);
  lcd.print(input_voltage); //print the voltage to LCD
  lcd.print(" V");
  lcd.setCursor(0,1);
  lcd.print("OUTPUT = LOW");
  if (input_voltage > 2.60) //set the voltage considered low battery here
  {
    lcd.setCursor(0,1);
    lcd.print("OUTPUT = HIGH");
  }
  delay(500);
}
```

Circuit Description

The measured battery or memory value is connected to an analog input (A0) on the Arduino through a voltage divider to ensure the input stays within 0-5V. A potentiometer adjusts the LCD contrast. If using an I2C backpack for the LCD, only SDA and SCL lines plus power (5V and GND) are required. On the protoboard, components such as an op-amp or buffer IC may be present to condition the signal depending on the sensor used.

Arduino Code Functionality

The Arduino sketch reads an analog value from A0 using `analogRead()`, converts it to a voltage using the reference (usually 5.0V) and the ADC resolution (1024), and then prints

the numeric voltage to the 16x2 LCD. The code also contains a threshold check to label the output as 'LOW' or 'HIGH' based on a defined voltage threshold, which can be used to indicate low battery. The LCD is initialized with the I2C address (for example 0x27) when using an I2C interface.

Methodology

- Assemble the components on the protoboard and wire the Arduino, LCD, battery snap, and voltage divider.
- Upload the Arduino sketch that reads A0, converts the ADC reading to voltage, and prints to the LCD.
- Adjust potentiometers for stable LCD contrast and fine-tune the voltage divider if necessary.
- Test with known voltages (or the battery) and verify correct readings and threshold behavior.

Expected Results and Applications

The system should display the measured voltage in real-time and indicate a 'LOW' or 'HIGH' status based on the configured threshold. This simple platform can be extended to read stored memory values from sensors, perform data logging to an SD card, or transmit readings over serial/Bluetooth for remote monitoring.

Conclusion

This project provides a hands-on introduction to analog sensing, ADC conversion, and LCD display interfacing with an Arduino. It is suitable for educational demonstrations and for building small monitoring devices such as battery testers or simple data-loggers.